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S26 Internetworking of things  
Lab 4  
03/23/26

P1)

```
Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
IPv4 Address. . . . . : 192.168.1.116
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1

C:\Windows\System32>arp -d

C:\Windows\System32>arp -a

Interface: 192.168.56.1 --- 0xd
Internet Address      Physical Address      Type
224.0.0.22           01-00-5e-00-00-16    static

Interface: 192.168.1.116 --- 0x11
Internet Address      Physical Address      Type
192.168.1.1          70-a7-41-d9-0d-07    dynamic
224.0.0.22           01-00-5e-00-00-16    static

C:\Windows\System32>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=1ms TTL=64
Reply from 192.168.1.1: bytes=32 time=3ms TTL=64

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms

C:\Windows\System32>arp -a

Interface: 192.168.56.1 --- 0xd
Internet Address      Physical Address      Type
224.0.0.22           01-00-5e-00-00-16    static
224.0.0.251          01-00-5e-00-00-fb    static

Interface: 192.168.1.116 --- 0x11
Internet Address      Physical Address      Type
192.168.1.1          70-a7-41-d9-0d-07    dynamic
224.0.0.22           01-00-5e-00-00-16    static
224.0.0.251          01-00-5e-00-00-fb    static

C:\Windows\System32>
```

```
C:\Windows\System32>getmac

Physical Address      Transport Name
-----
N/A                  Hardware not present
34-0E-DE-6D-96-FB    \Device\NPF{93099C4B-5491-4E16-9980-7399BA04154C}
D8-5E-D3-0E-E9-74    Media disconnected
0A-00-27-00-00-0D    \Device\NPF{468F1293-02BC-4339-A6C4-262CFC5BA249}
```

| No. | Time        | Source            | Destination    | Protocol | Length | Info                                    |
|-----|-------------|-------------------|----------------|----------|--------|-----------------------------------------|
| 274 | 8.642378000 | Intel_6d:96:fb    | Broadcast      | ARP      | 42     | Who has 192.168.2.1? Tell 192.168.2.169 |
| 275 | 8.643626600 | Ubiquiti_d9:0d:07 | Intel_6d:96:fb | ARP      | 60     | 192.168.2.1 is at 70:a7:41:d9:0d:07     |

|                                                                                                    |      |                                                 |
|----------------------------------------------------------------------------------------------------|------|-------------------------------------------------|
| ▶ Frame 274: Packet, 42 bytes on wire (336 bits), 42 bytes captured on interface 0                 | 0000 | ff ff ff ff ff ff b4 0e de 6d 96 fb 08 06 00 01 |
| ▼ Ethernet II, Src: Intel_6d:96:fb (b4:0e:de:6d:96:fb), Dst: Broadcast (ff:ff:ff:ff:ff:ff)         | 0010 | 08 00 06 04 00 01 b4 0e de 6d 96 fb c0 a8 02 a9 |
| ▶ Destination: Broadcast (ff:ff:ff:ff:ff:ff)                                                       | 0020 | 00 00 00 00 00 00 c0 a8 02 01                   |
| ▶ Source: Intel_6d:96:fb (b4:0e:de:6d:96:fb)                                                       |      |                                                 |
| Type: ARP (0x0806)                                                                                 |      |                                                 |
| [Stream index: 2]                                                                                  |      |                                                 |
| ▼ Address Resolution Protocol (request)                                                            |      |                                                 |
| Hardware type: Ethernet (1)                                                                        |      |                                                 |
| Protocol type: IPv4 (0x0800)                                                                       |      |                                                 |
| Hardware size: 6                                                                                   |      |                                                 |
| Protocol size: 4                                                                                   |      |                                                 |
| Opcode: request (1)                                                                                |      |                                                 |
| Sender MAC address: Intel_6d:96:fb (b4:0e:de:6d:96:fb)                                             |      |                                                 |
| Sender IP address: 192.168.2.169                                                                   |      |                                                 |
| Target MAC address: 00:00:00_00:00:00 (00:00:00:00:00:00)                                          |      |                                                 |
| Target IP address: 192.168.2.1                                                                     |      |                                                 |
| ▶ Frame 275: Packet, 60 bytes on wire (480 bits), 60 bytes captured on interface 0                 | 0000 | b4 0e de 6d 96 fb 70 a7 41 d9 0d 07 81 00 00 00 |
| ▼ Ethernet II, Src: Ubiquiti_d9:0d:07 (70:a7:41:d9:0d:07), Dst: Intel_6d:96:fb (b4:0e:de:6d:96:fb) | 0010 | 08 06 00 01 08 00 06 04 00 02 70 a7 41 d9 0d 07 |
| ▶ Destination: Intel_6d:96:fb (b4:0e:de:6d:96:fb)                                                  | 0020 | c0 a8 02 01 b4 0e de 6d 96 fb c0 a8 02 a9 00 00 |
| ▶ Source: Ubiquiti_d9:0d:07 (70:a7:41:d9:0d:07)                                                    | 0030 | 00 00 00 00 00 00 00 00 33 50 fb c5             |
| Type: 802.1Q Virtual LAN (0x8100)                                                                  |      |                                                 |
| [Stream index: 1]                                                                                  |      |                                                 |
| ▶ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 0                                                        |      |                                                 |
| ▼ Address Resolution Protocol (reply)                                                              |      |                                                 |
| Hardware type: Ethernet (1)                                                                        |      |                                                 |
| Protocol type: IPv4 (0x0800)                                                                       |      |                                                 |
| Hardware size: 6                                                                                   |      |                                                 |
| Protocol size: 4                                                                                   |      |                                                 |
| Opcode: reply (2)                                                                                  |      |                                                 |
| Sender MAC address: Ubiquiti_d9:0d:07 (70:a7:41:d9:0d:07)                                          |      |                                                 |
| Sender IP address: 192.168.2.1                                                                     |      |                                                 |
| Target MAC address: Intel_6d:96:fb (b4:0e:de:6d:96:fb)                                             |      |                                                 |
| Target IP address: 192.168.2.169                                                                   |      |                                                 |

1)

No, it's what's used in IPV4, but it's been replaced by NDP in IPV6

2)

| From                           |               | To                             |               |                                     |
|--------------------------------|---------------|--------------------------------|---------------|-------------------------------------|
| MAC                            | IP            | MAC                            | IP            | ARP Packet Length                   |
| B4:0e:de:6d:96:fb<br>Intel_    | 192.168.2.169 | 00:00:00:00:00:00<br>Broadcast | 192.168.2.1   | 42 bytes<br>14eth, 28 arp           |
| 70:a7:41:d9:0d:07<br>Ubiquiti_ | 192.168.2.1   | B4:0e:de:6d:96:fb<br>Intel_    | 192.168.2.169 | 60 bytes<br>14 eth, 18 vlan, 28 arp |

Note: I got these results on the guest wifi, switching to our primary network resulted in a 22byte vlan, and a 64 byte reply packet

3)

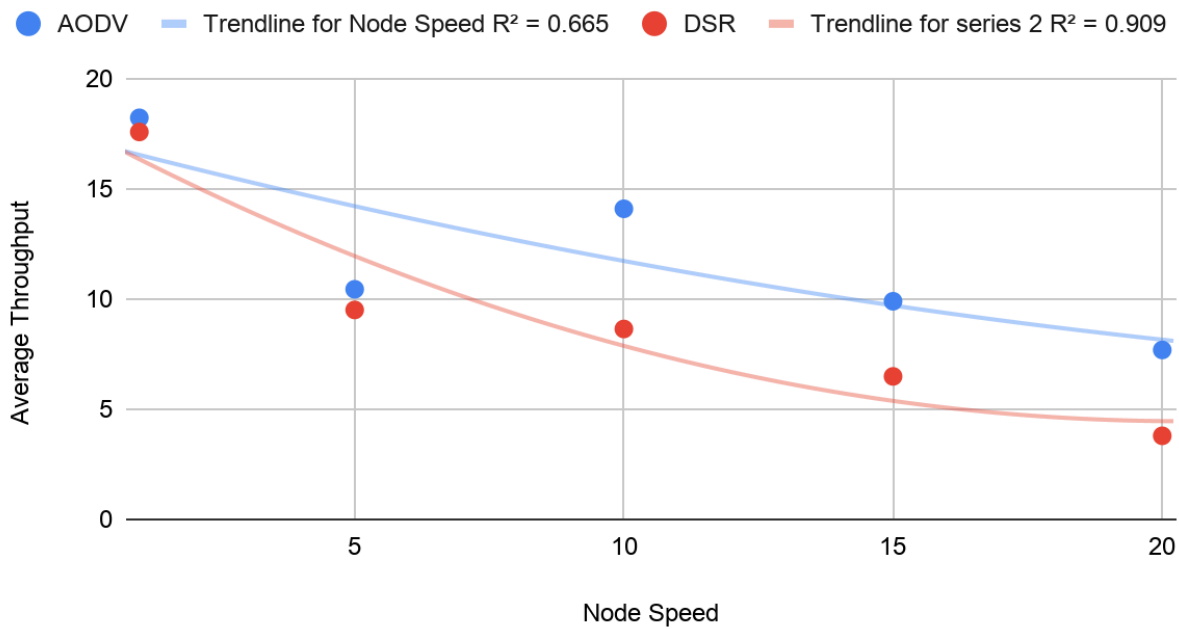
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P2)

AODV broadcasts and uses routing tables at each node while DSR uses source routing, finding a whole route and storing it in the packet header before sending.

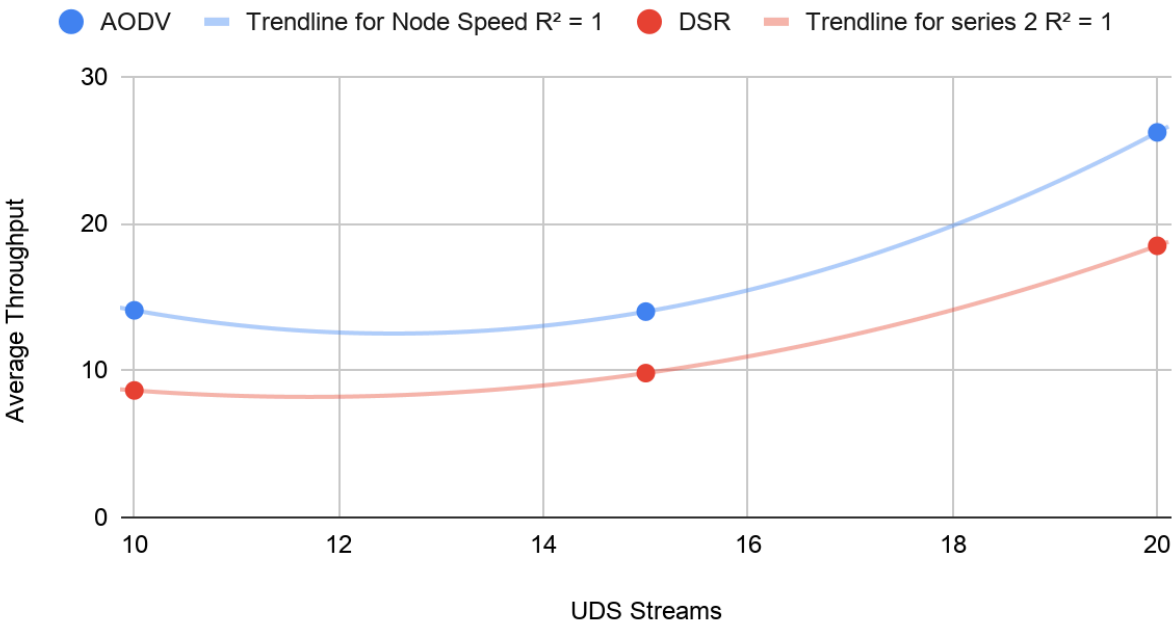
| Node Speed |             |             |
|------------|-------------|-------------|
|            | AODV        | DSR         |
| 1          | 18.21478788 | 17.57349495 |
| 5          | 10.43652525 | 9.500444444 |
| 10         | 14.08775758 | 8.636767677 |
| 15         | 9.893494949 | 6.485333333 |
| 20         | 7.685171717 | 3.790868687 |

## Node Speed



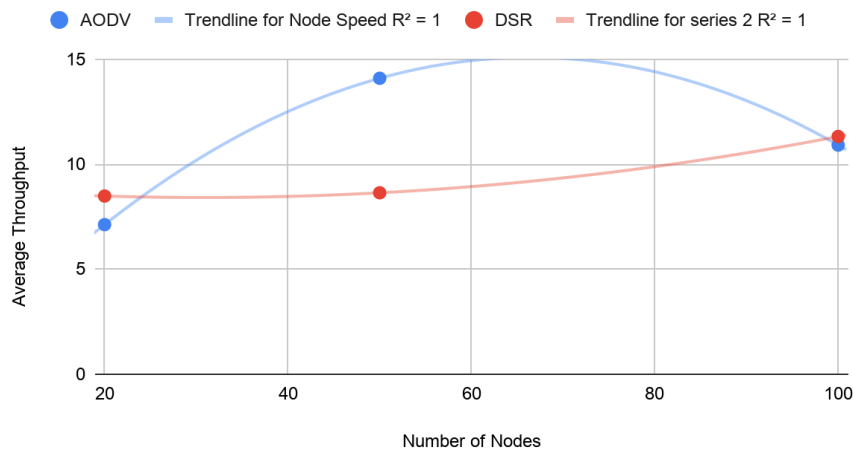
| UDP Streams |             |             |
|-------------|-------------|-------------|
|             | AODV        | DSR         |
| 10          | 14.08775758 | 8.636767677 |
| 15          | 13.99983838 | 9.815919192 |
| 20          | 26.19474747 | 18.47854545 |

## UDP Streams



| Num Nodes |             |             |
|-----------|-------------|-------------|
|           | AODV        | DSR         |
| 20        | 7.121454545 | 8.481616162 |
| 50        | 14.08775758 | 8.636767677 |
| 100       | 10.90715152 | 11.31571717 |

Number of Nodes



#### 6. Discuss:

Throughput is not a linear function of node speed as with increasing speed you get increasing failures and overhead, decreasing throughput. It's hard to determine the best shape of this plot with so few points and, for example, AODV speed=5 being significantly slower than both 1 and 10, but we would generally expect the dropoff to get sharper at sufficiently high speeds.

#### 7. Recommendation:

AODV consistently performs at least slightly better than DSR in our first two tests, but falls slightly short in our Number of Nodes test. In test 1 AODV will perform better relative to DSR with increasing mobility since it's able to adapt faster, which we see in the increasing gap between the two. In test 2 ADOV will perform better with per-destination routing tables while DSR gets per-packet overhead.

#### 8. 6660

In test 3, we see interesting results. At 50 nodes, ADOV is faster for all the reasons mentioned above, but for 20 nodes, these optimizations are unnecessary and just cause overhead, while DSR is able to thrive, and for 100 nodes, the overhead of AODV RREQs becomes massive, allowing DSR to once again take over.